ADMINISTRATIVE CAPITAL FOR URBAN DEVELOPMENT
العاصمة الادارية للتنمية العمرانية

الرواد الرقميون
DiGiLiANS

SpaceEye GeoAI

New Capital Radar: a company that owns a city.

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Problem in numbers

>400
developers

>500
projects

3 months
between field reviews

7-8%
of developers seriously late

6
mutual terminations

More than 500 projects, and a field review every 3 months → Which compounds come first? → A short cycle list, with reasons

- A committee cannot visit everything every cycle: whom to visit first? These are the chairman's statements, not our measurement.
- The gap: ACUD's data (plot boundaries, handover dates, supervisor reports, field review records) is what would turn our public-data screening into operation.

Source: statements by the ACUD chairman, as recorded in docs/requirements/01-specs.md (section 12, slide 1 and the data request list).

The question and the decision

Today (as-is): every one of the 12 compounds gets a field review every 3 months, 4 visits a year each.

Tomorrow (to-be): a cycle list of at most 4 compounds from levels 1-3, each with a reason sentence. The committee decides.

12 R7 compounds in scope

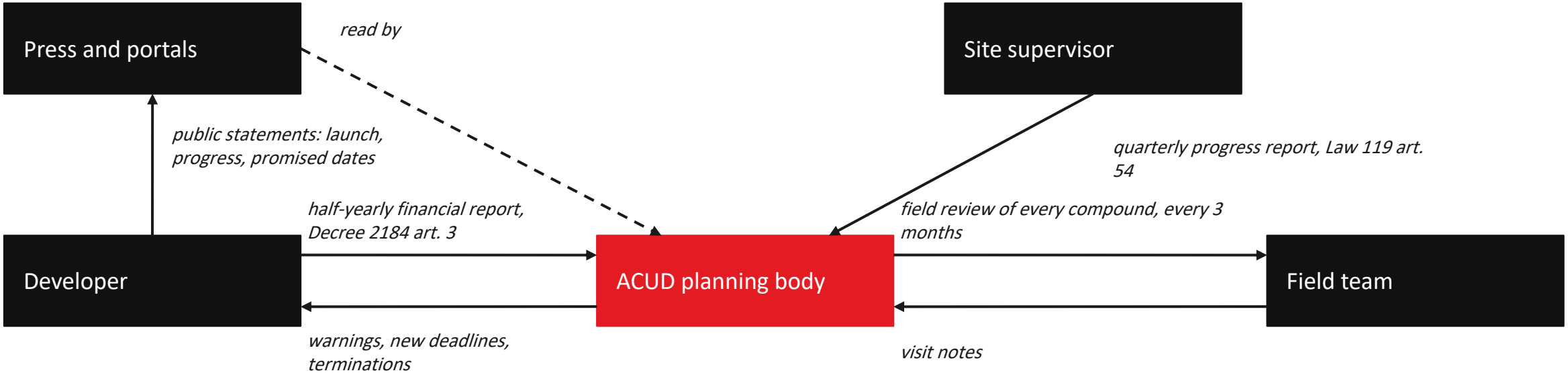
4 visits per cycle

Capacity 4 of 12 compounds, from levels 1-3

- Question: which R7 compounds should the committee visit first next cycle, and why?
- The observatory ranks; the committee decides.
- Scope: R7 only, 12 coded compounds, reporting quarter 2026Q3, cycle capacity 4.
- ACUD has not given us visit times or missed cases, so the as-is pain is not measured.
- Wording rule: the list says "no building activity since X" or "passed its first date by Y", never a label on a developer.
- Frame: preventive · Buildings & Facilities · normal conditions · stability. Affected: buyers (6,474 homes in the five late public cases). Why now: C18 is 57 months past its first announced handover date.

Source: configs/cost_assumptions.yaml (projects: 12, visits_per_year: 4), docs/requirements/01-specs.md (sections 0, 2, 7 and 12); docs/deliverables/decision-memo.md (6,474 homes, why now).

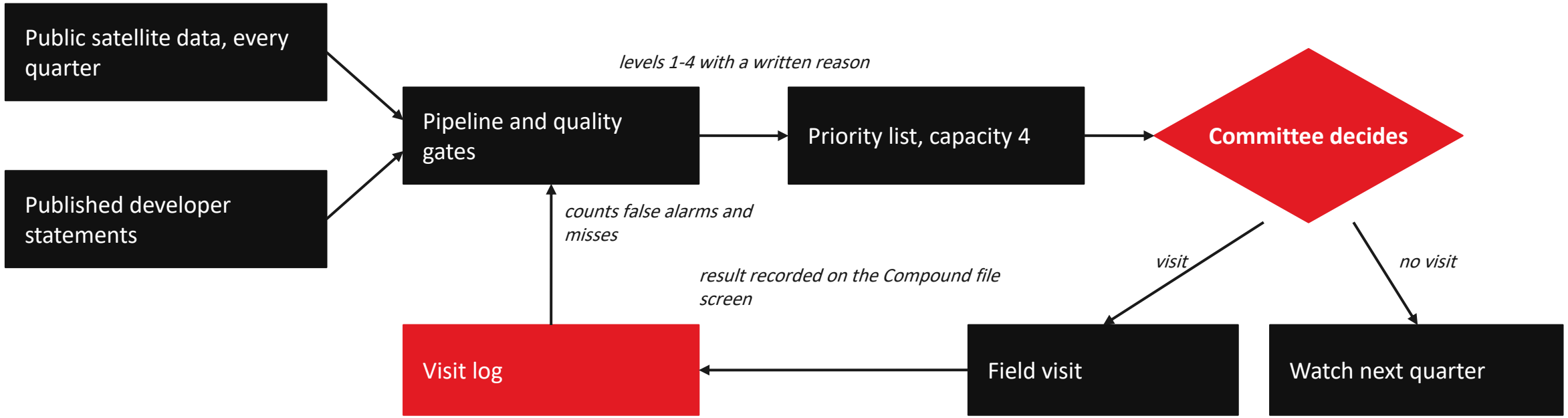
As-is: how R7 is followed today



- Today: the same field review every 3 months for every compound, 48 reviews a year in R7 (12 x 4). 5 of the 6 public late cases ran 18 to 33 months past the original promise (median 25).
- Three causes: effort is not matched to risk; progress comes from the developer's own statements; nobody compares pace with the promised date.
- Evidence: C19 claimed 70% while 28% of its built area was 20 m or taller that year (a reason to look, not proof); C14 reported 35% three months before its promised date and moved it; activity ranges 0.03 to 0.90 around a 0.61 mean.
- Rebuilt from public sources; ACUD's workflow was not observed and must be confirmed.

Source: docs/deliverables/as-is.md (process map, baseline, root causes).

To-be: the new process



- Roles: the Track A owner runs the pipeline and gates each quarter (second reviewer); the committee decides; the field visit lead records the result; the ACUD data custodian keeps the code-to-name mapping.
- Targets (proposed): every level 1-2 compound visited (31 of 31 quarters covered); 12 of 12 results recorded.
- ISO 37122 and U4SSC have no matching project-level indicator; targets rest on R7 data. S3 tests load above capacity.
- Proposed by the team, not approved by ACUD. The system recommends; the committee decides.

Source: docs/deliverables/to-be.md (the new process, roles and proposed targets).

From Data Scarcity to Data Intelligence

How can we monitor construction without ACUD's internal developer records?

1 · OBSERVE

Satellite imagery and land-cover change

What is physically changing on site?

public imagery → derived signals

2 · CONTEXTUALIZE

Published dates, announcements and regulations

What was publicly promised, and when?

real, with source and publish date

3 · SIMULATE

Synthetic finance and operations records

How would the system behave with ACUD's own data?

generated · training and testing only

Fused per compound and quarter → Track A. An announcement is context, never field proof.

Track B only, every number marked ●

12
compounds in R7

43
quarters, 2016Q1 to 2026Q3

516
compound-quarter observations

57
public statements, each sourced

Source: app_data/real (compounds 12, sat_quarterly 516 rows = 12 x 43 quarters, statements 57); SPACEEYE_DATA_REFERENCE.md sections 4 and 13.

From Pixels to Construction Signals

1 · Define the site
Compound boundary polygon

2 · Measure each quarter
Dynamic World + Sentinel-2

3 · Estimate activity
Gradient boosting → score A

4 · Check by eye
Dated Esri Wayback before / after

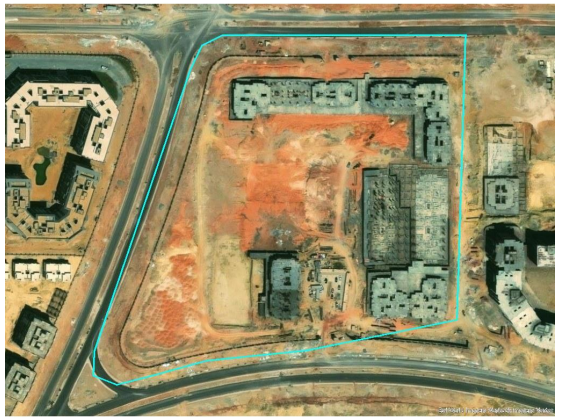
C07 | capture 2024-05-15



Wayback release 2024-10-10 | Esri World Imagery Wayback; imagery: Maxar. Service credits: Esri, Vantor, Earthstar Geographics, and the GIS User Community
Cyan: reviewed working boundary, not a legal survey. Source dates may vary within the map.
Static educational report figure. Third-party imagery rights retained; not an open imagery dataset.

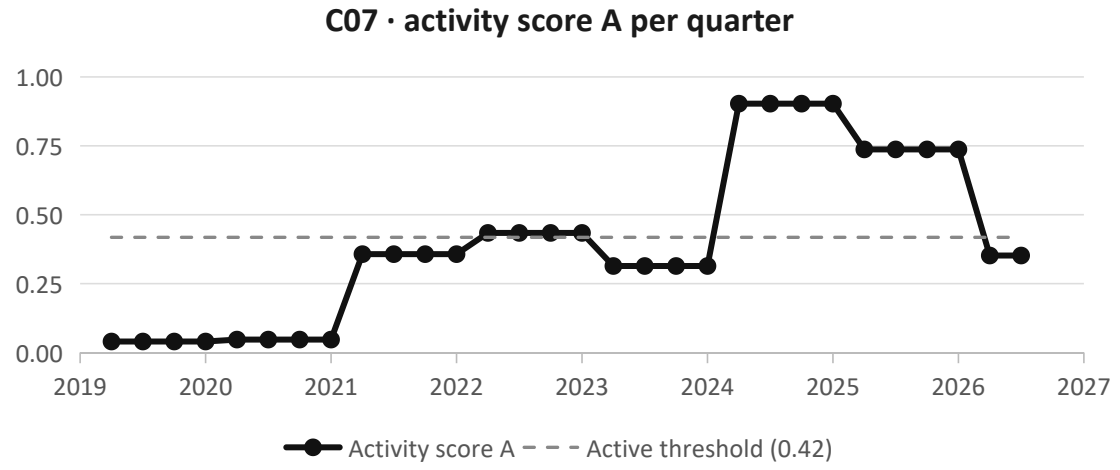
C07 · *Before · 2024-05-15*

C07 | capture 2025-07-23



Wayback release 2026-05-28 | Esri World Imagery Wayback; imagery: Vantor. Service credits: Esri, Vantor, Earthstar Geographics, and the GIS User Community
Cyan: reviewed working boundary, not a legal survey. Source dates may vary within the map.
Static educational report figure. Third-party imagery rights retained; not an open imagery dataset.

C07 · *After · 2025-07-23*



2026Q2-Q3: below the threshold after the start → K2 = 2

Satellite signals indicate activity and change, not verified completion or handover.

Source: app_data/real/sat_quarterly.parquet (C07 activity score A); model/activity_meta.json (threshold 0.42); Esri World Imagery Wayback (imagery: Maxar 2024-05-15, Vantor 2025-07-23; Esri, Vantor, Earthstar Geographics, GIS User Community).

From Evidence to Inspection Priorities

Three questions. One transparent decision.

Has it started?
K6 · time since launch or promise, no observed start

Has it stopped?
K2 · consecutive inactive quarters after the start

Is it past its announced date?
K3 · months past the first published handover date

Explainable rule
The most urgent condition that applies sets the level. No weights, no black box.

Safeguard: "may be complete" holds it at level 3 until a site check.

1 **Inspect now**

2 **Inspect this cycle**

3 **Document review**

4 **Routine**

Real examples · 2026Q3

C18 → Level 1
Still active (K2 = 0), yet 57 months past its first announced handover date (Dec 2021).

C07 → Level 2
2 inactive quarters in a row, with no published handover date at all.

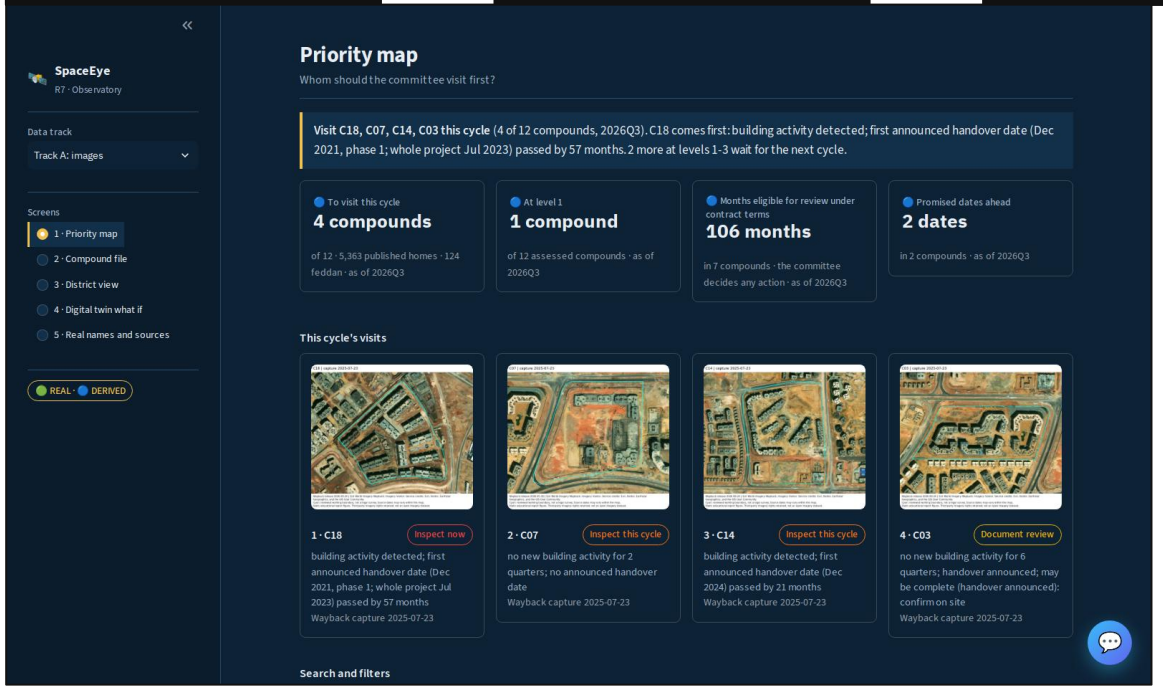
C03 → Level 3
6 inactive quarters, but a handover was announced: may be complete, confirm on site.

The system prioritizes inspections. The committee makes the decision.

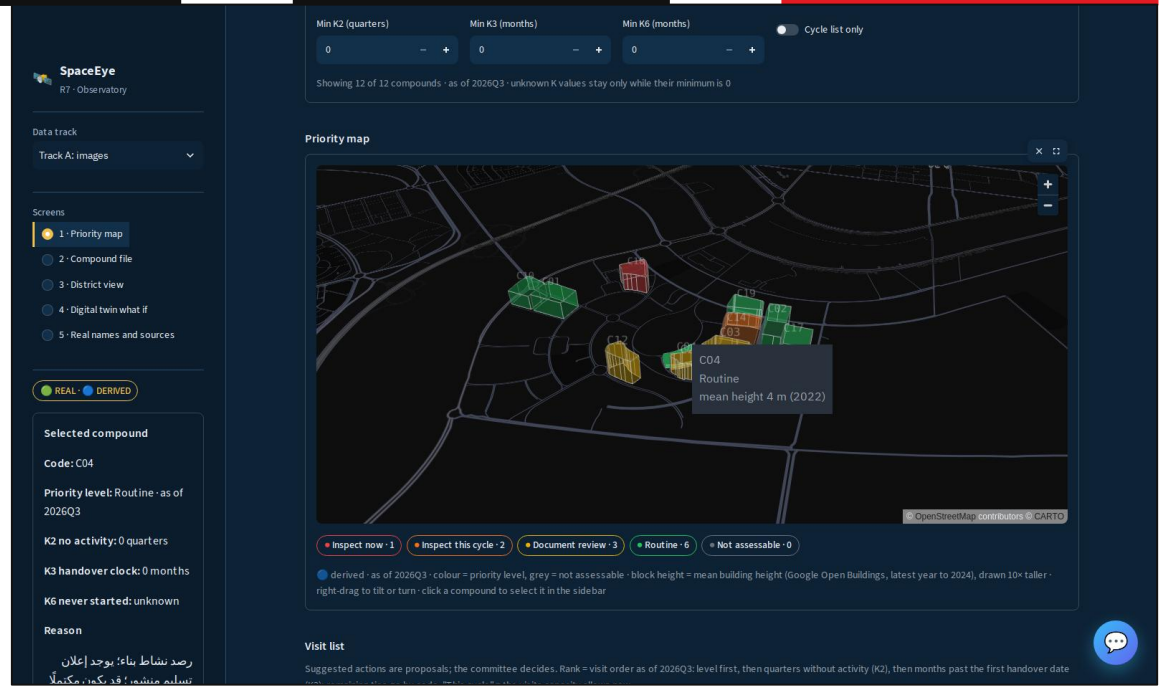
Source: configs/data.yaml (thresholds fixed before any score was seen); app_data/real/kpi_values.parquet (levels and reasons at 2026Q3).

Live demo (Track A)

Priority map → Cycle list (capacity 4, levels 1-3) → Compound reason sentence → Inspection file → Visit result recorded



Dashboard screenshot · Priority map

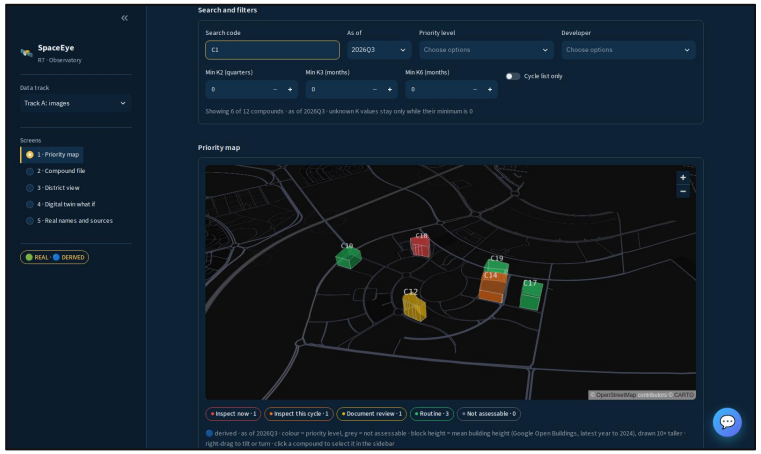


Dashboard screenshot · Priority map, selected compound

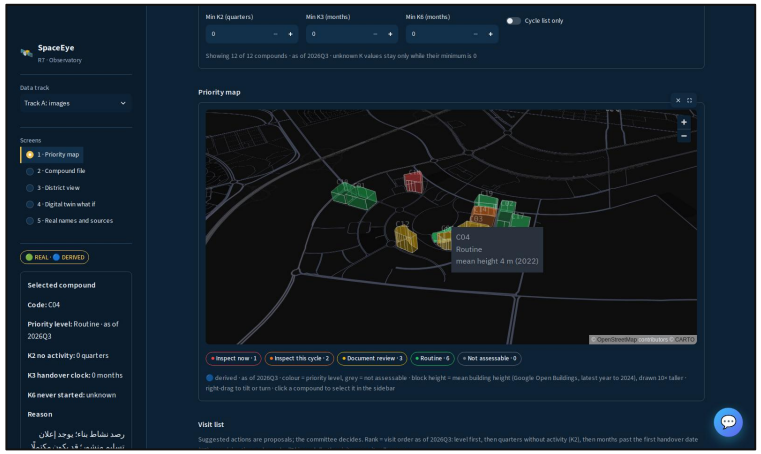
- Track A on screen is real or derived, no generated number on the map. Run: make run (T-AH-19), from stored files.

Source: docs/deliverables/user-guide.md; board task T-AH-19 in docs/requirements/02-team-roles-and-tasks.md.

Search, filters and map click



Dashboard screenshot · Priority map, search and filters



Dashboard screenshot · Priority map, selected compound

- One search box: a compound code (C..) or developer code (D..); a prefix works. "C1" shows 6 of 12 compounds.
- Filters: as-of quarter, priority level, developer, minimum K2 / K3 / K6, cycle list only.
- Rank stays the quarter's visit order when you filter.
- Click a block on the map: the sidebar shows that compound. "Clear selection" resets. Codes only.

Source: app.py (render_filters, open_clicked_compound, render_selected_compound_panel), src/dashboard/insights.py (filter_compounds); board tasks T-NA-11, T-NA-12, T-NA-13 and T-EL-21 in docs/requirements/02-team-roles-and-tasks.md.

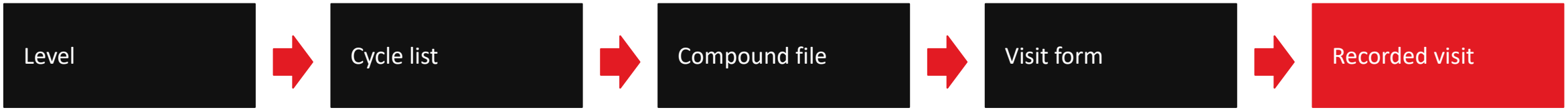
Four screens, one path from alert to action

1 · Priority map
Whom first?

2 · Compound file
Why this one?

3 · District view
What does the system give, and when does it fail?

4 · Digital twin what if
Who changes if costs, collections or capacity change? Track B, generated



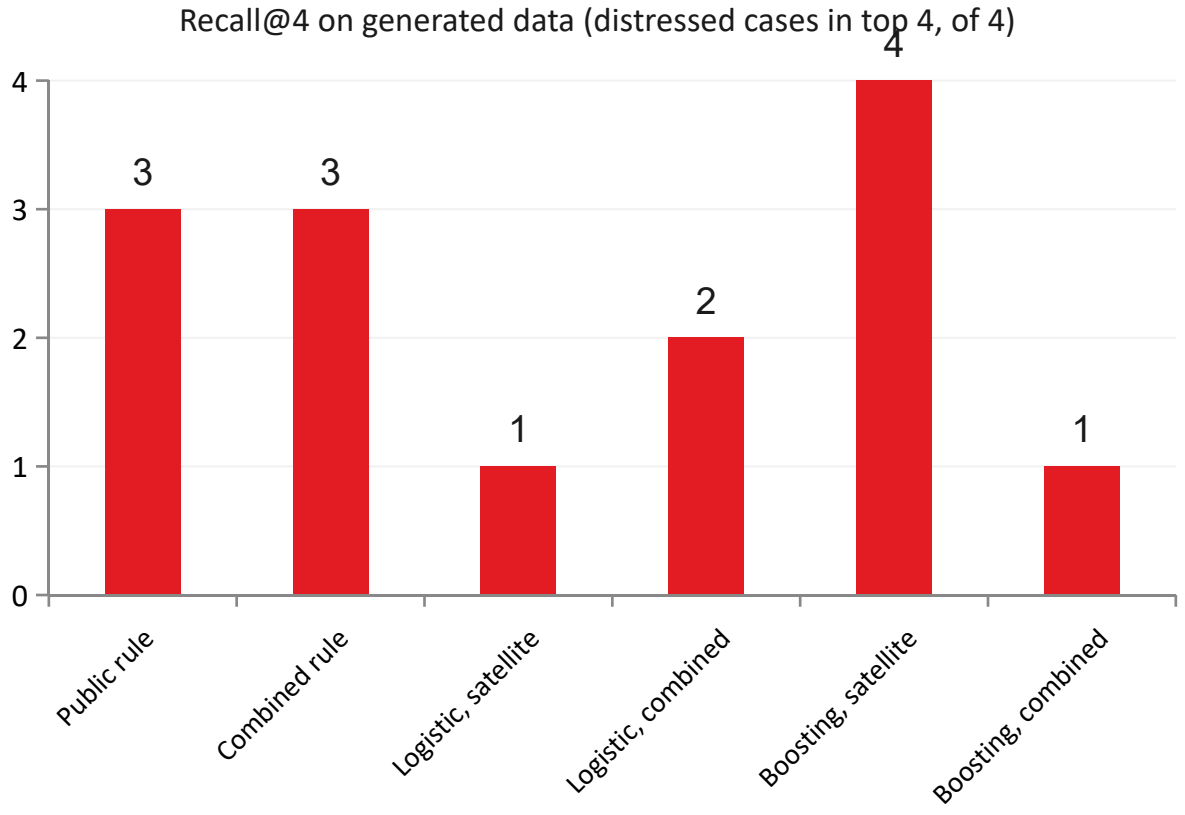
- Alert to action: the level sets the cycle list, the compound file gives the reason, the visit form records the result.
- Audience test: the teammate reviews changed the screens (audience-test.md). The supervisor took an unrecorded ten-second test; a recorded one with a committee member is still to be run.

Source: docs/deliverables/user-guide.md (screen subtitles), docs/deliverables/audience-test.md, docs/deliverables/display-rules-check.md.

Track B: generated data

Synthetic - for training only

- Target: distressed / not_distressed in the next six months. Completion prediction is out of scope.
- 96 training rows, 48 test rows; only 4 test rows are distressed, so power is very low.
- Boosting 4 of 4 rests partly on height; without height (v2a) it is 1 of 4.
- Random top-4 baseline: mean 0.335, 95% interval 0.000 to 0.750 (2,000 draws).
- With 4 cases the methods cannot be separated, and combining ACUD-style data did not clearly help. Scores are diagnostic, not real-world accuracy.

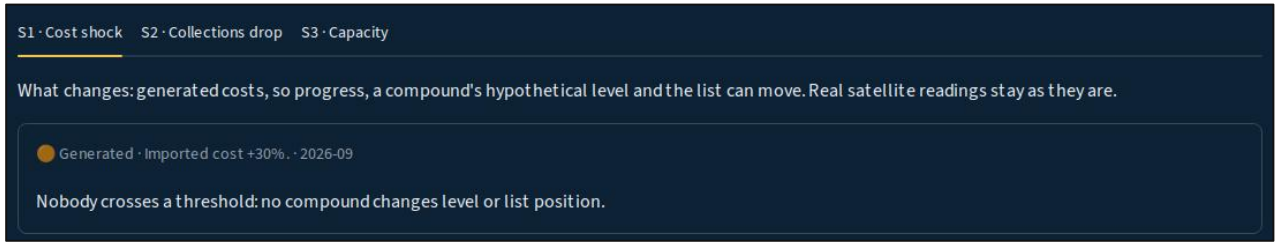


Source: docs/deliverables/complaints_eval.csv, data/EW_V2A_VS_V1.md.

The observatory: one what-if scenario

Synthetic - for training only

Scenario	What changes	Rows that change list membership
S1	Dollar shock	0 of 216; levels never change
S2	Collections drop	0 of 216; levels never change
S3	Inspection capacity 2 or 6 instead of 4	45 of 744 (28 at capacity 2, 17 at capacity 6); levels never change



Dashboard screenshot · Digital twin what if

- Light compound-level observatory: observed state from satellites and statements, simulated hidden state from the generator.
- Demo S3 live. Say plainly that S1 and S2 change nothing in the list or levels.
- Orange banner on screen, no map. Not a 3D or real-time observatory.

Source: logs/review-05.md (scenario row counts).

Ask the project: the chatbot

Question → Intent router (10 intents) → Hybrid retrieval: DuckDB facts + Chroma search → Answer: Gemini if approved and within budget, else local → Sources shown

- Opens from the dashboard's 💬 button (port 8001).
- Follow-up questions keep the last compound code.
- Generated items stay labelled "Synthetic - for training only": complaints and visit records are synthetic.
- Retrieval reads the v7 release facts and the statements, complaints and knowledge documents.



Dashboard screenshot · Chatbot, start screen

Source: chatbot/app/retrieval/router.py (intents), chatbot/app/retrieval/hybrid.py, chatbot/app/api/main.py (session context), chatbot/app/generation/gemini_client.py, configs/ai.yaml, README.md.

What the chatbot knows, and its limits

8 of 8

intent tests passed (2026-11-03)

5 of 5

retrieval checks passed

Call budget: 200 model calls in total (approved by the project owner on 2026-11-04), then local answers

- Knowledge: 9 sourced documents (ACUD site, Decree 2184/2022, Building Law 119/2008 art. 54, Law 151/2020, Dynamic World, Open Buildings, Esri Wayback, ISO 37120/37122, team SLA note) plus the project's data.
- The tests are the team's own and small. They are not an accuracy claim.
- Model calls only with approval in configs/ai.yaml (chatbot.approval: approved), then local answers.
- Codes only. Not legal advice.

Source: chatbot/knowledge/SOURCES.csv, chatbot/evaluation_results.json, chatbot/retrieval_evaluation_results.json, configs/ai.yaml.

Risks and responsible use

A level is not a finding

It says whom to visit first; it is not misconduct, legal distress, breach or a verified fine. No automatic sanction.

Owner: Track A owner

Unknown stays unknown

Missing evidence is never read as negative. K3 relies on developer statements, not independent measurement.

Owner: Track A owner

Generated data

Never mixed with real data; Track B is marked and hidden from the map. It shows nothing about real accuracy or fairness.

Owner: Track B / AI owner

Small samples

6 real cases and 4 test cases: counts only, no percentages.

Owner: Field visit lead

Complaint text

Masking does not prove every real name is removed; real text needs further review. The chatbot sees only generated complaints.

Law 151/2020

Interpretation needs the data custodian's review. Not legal advice.

Owner: ACUD data custodian

Handover

Guides, method card and checklist are in the repo; no one outside the team has rebuilt the system from them yet.

The committee decides

Every visit result is recorded to build real ground truth.

Owner: Committee

Each risk has an owner role and an early sign in governance.md (proposed, not signed by ACUD).

Source: docs/deliverables/responsible-ai.md, docs/requirements/01-specs.md (Law 151/2020). | (knowledge transfer) docs/deliverables/technical-track-a.md, docs/deliverables/user-guide.md, docs/deliverables/knowledge-transfer.md, docs/deliverables/method-card.md, docs/deliverables/final-checklist.md.